Student Attendance System

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Project Title: Student Attendance System

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1.INTRODUCTION:

The Student Attendance System is designed to simplify the process of recording student attendance in educational institutions. Paper-work based attendance registers are time-consuming and difficult to maintain especially for multiple subjects and classes.

The system supports multiple subjects, each taught by a specific teacher with secure login Id and password. Teachers can log in, select their class and batch, and mark attendance for every student in their subject. Additionally, teachers can view all subject attendance, generate student-wise reports across all subjects, and also edit previously marked attendance to correct mistakes.

2.OBJECTIVES:

Digitalize the attendance process and allow teachers to maintain subject-wise attendance.

Store attendance for each student for the last 3 days.

Provide an accurate percentage-based attendance report.

Offer editing and viewing option, and simplify management of multiple classes and batches.

3.FEATURES

1. Teacher Login System

Each subject has a fixed teacher with a unique ID and password.

Secure login authentication before accessing attendance.

2. Class & Batch Section

Teacher can select between 11th and 12th commerce classes.

Each class has two batches (A and B).

3. Student Record

Each batch has pre-defined students.

Attendance is stored separately for each class and batch.

4. Attendance Recording

Teacher mark each student as P(Present) and A(Absent).

System automatically maintains a 3-days rolling history.

5. View Attendance

View all student's attendance for the last 3 days.

Calculate and display the overall class attendance percentage.

View attendance of specific student with individual percentage.

Date and time is display when the record is saved.

6.Edit Attendance

Teacher can correct mistakes by editing any of three days.

4.WORKFLOW OF SYSTEM

1.Start Program

↓

2.Teacher select subject

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3.Login with teacher ID and Password

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4.Select class (11th/12th)

↓

5.Select Batch (A/B)

↓

5. Attendance Menu

Mark today ‘s attendance

View all attendance

View specific student

Edit attendance

Exit

6.Attendance is recorded and stored automatically.

5.OUTPUT

```
--- Select Subject ---
1. Accountancy
2. Business Studies
3. Economics
4. Mathematics
Enter choice: 1

--- LOGIN (Accountancy) ---
Teacher ID: acc123
Password: acc@123
Welcome priya-Mam

--- Select Class ---
1. 11th Commerce
2. 12th Commerce
Enter class number: 1

--- Select Batch ---
A. Batch A
B. Batch B
Enter batch: A

--- MENU ---
1. Mark Attendance
2. View All
3. View Student
4. Edit
5. Exit
Choice: 1

--- Mark Attendance ---
```

```
--- Mark Attendance ---
Payal (P/A):
Saniya (P/A): p
Priya (P/A): p
Vishu (P/A): p
Aniket (P/A): p
 Attendance Marked with Time & Day

--- MENU ---
1. Mark Attendance
2. View All
3. View Student
4. Edit
5. Exit
Choice: 2

--- Attendance (Last 3 Days) ---
Payal: ['A', 'P', 'A', '2025-12-12 07:37 PM'] | 33.33% | FAIL
Saniya: ['A', 'P', 'P', '2025-12-12 07:37 PM'] | 66.67% | PASS
Priya: ['A', 'P', 'P', '2025-12-12 07:37 PM'] | 66.67% | PASS
Vishu: ['A', 'P', 'P', '2025-12-12 07:37 PM'] | 66.67% | PASS
Aniket: ['A', 'P', 'P', '2025-12-12 07:37 PM'] | 66.67% | PASS

 Class Attendance = 60.00%

--- MENU ---
1. Mark Attendance
2. View All
3. View Student
4. Edit
5. Exit
Choice: 3
```

```
Choice: 3

Students: ['Payal', 'Saniya', 'Priya', 'Vishu', 'Aniket']
Enter name: Payal

Payal: ['A', 'P', 'A', '2025-12-12 07:37 PM']
Attendance = 0.00%
Status = FAIL

--- MENU ---
1. Mark Attendance
2. View All
3. View Student
4. Edit
5. Exit
Choice: 4

--- Edit Attendance ---
Students: ['Payal', 'Saniya', 'Priya', 'Vishu', 'Aniket']
Enter name: Payal
Current: ['A', 'P', 'A', '2025-12-12 07:37 PM']
Day to edit (1/2/3): 1
New value (P/A): P
Updated (With New Time)

--- MENU ---
1. Mark Attendance
2. View All
3. View Student
4. Edit
5. Exit
Choice: 5
```

6.ADVANTAGES

- Easy to use for teachers
- Reduce paper work
- Tracks attendance trends quickly
- Allows editing mistakes
- Maintain recent attendance history automatically

7.RELATIONSHIPS

- **Teacher ↔ Subject**

  Type: **One - To - One**

  Each subject has one fixed teacher

  Each Teacher teaches only one subject

- **Subject ↔ Class & Batch**

  Type: **One - To - Many**

  It means one subject can be taught different class and batches

- **Class & Batches ↔ Student**

  Type: **One - To - Many**

  One class and batch contain many students

- **Student ↔ Attendance**

  Type: **One - To – Many**

  One student has multiple attendance records

- **Attendance ⟷ Subject**

  Type: **One - To - Many**

One attendance system is managed by multiple subjects.

8.CONCLUSION

The student attendance system simplifies the attendance process through automation. This dataset helps in managing student attendance in an organized and systematic way. It clearly stores information about teachers, subjects, student groups, dates, and attendance status. Using this dataset reduces manual work, avoids errors, and makes attendance tracking faster and more accurate. Overall, it improves efficiency and helps teachers maintain proper attendance records easily.

9.FUTURE ENHANCEMENT

Add student and parent login features.

Send automatic alerts to parents for low attendance.

Notify students when attendance is marked absent.

Use biometric fingerprint RFID cards.